

MSIN0097 Predictive Analytics

Individual Assessment

Candidate Number: VSXB6

Word Count: 1863 words

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1. Introduction

In digital marketing, companies allocate budgets across various channels, including paid search, shopping campaigns, and social media. Marketers must determine how to allocate budgets efficiently to maximise revenue. Since performance changes across channels and over time, relying only on past spending may lead to inefficient decisions.

Predictive analytics uses data to estimate how marketing influences revenue. By modelling historical spend and purchasing behaviour, marketers can forecast revenue and see how investment changes affect performance.

This project develops an end-to-end predictive modelling workflow to analyse marketing channel effectiveness and support budget allocation decisions. Using historical marketing data, several forecasting models are trained and evaluated to predict revenue outcomes. The selected model is then used to simulate incremental spending scenarios across channels and estimate the expected revenue uplift. Based on these simulations, the analysis provides recommendations on which channels should receive additional investment and how marketing budget could be allocated to maximise revenue.

2. Problem Framing and Dataset

The objective of this project is to analyse how marketing expenditure across different channels influences purchasing outcomes and to use these insights to support marketing budget allocation decisions. The key business question is:

If an additional marketing budget becomes available, which channels should receive the investment, and what percentage should be allocated to each channel, in order to maximise revenue?

This problem is formulated as a supervised regression task to predict purchase outcomes from historical marketing activity.

The target variable used in this study is the total number of purchases, which acts as a proxy for revenue performance. The predictive features include marketing spend across

several digital channels, together with engagement indicators such as clicks. These variables capture the relationship between marketing investment and consumer purchasing behaviour.

Model performance is evaluated using regression metrics, including MAE, RMSE, and MAPE, while model interpretability is also considered when selecting the final approach. The analysis assumes that historical relationships between marketing spend and purchases remain relatively stable over time. However, external factors such as seasonality and market conditions are not explicitly modelled, which may affect prediction accuracy. Agent tooling (Codex) is used to assist with code generation and debugging, while all modelling choices and results are manually verified.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the dataset's structure, statistical relationships, and quality before building predictive models. The dataset contains daily observations of marketing performance variables, including advertising spend, user engagement metrics such as clicks and impressions, and purchasing outcomes. As the dataset is time-indexed, preserving chronological order is important when analysing trends and preparing the modelling pipeline.

Initial data quality checks revealed high missing rates in several media variables, primarily because certain channels were inactive on many days. These missing values were interpreted as zero activity and replaced with zeros during preprocessing. After this step, the processed dataset contained no missing values, ensuring consistency for subsequent modelling.

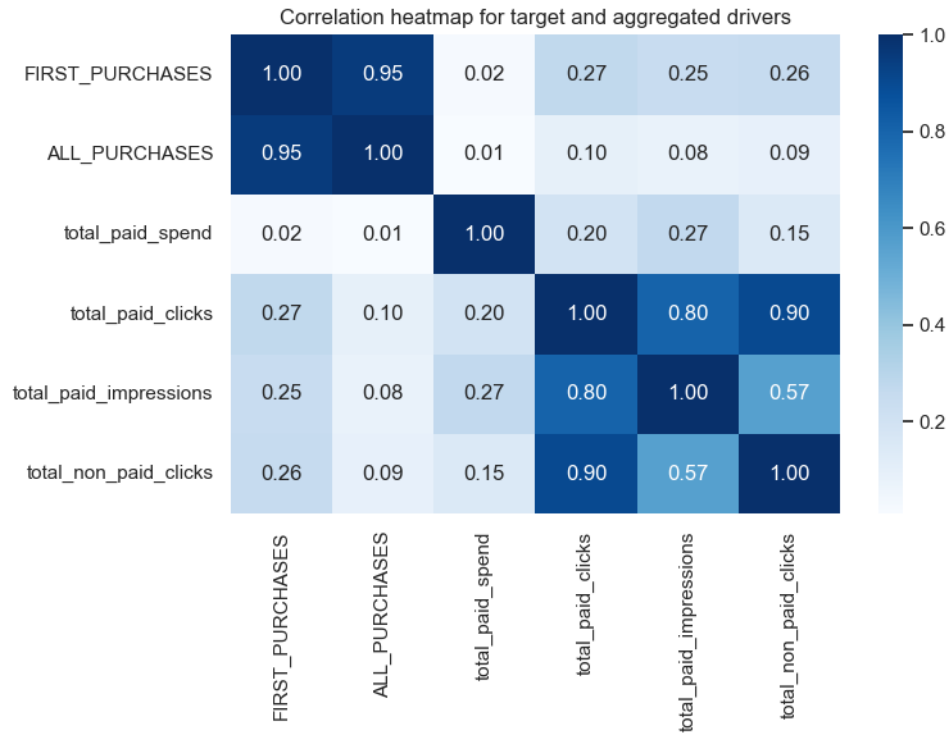


Figure 1. Correlation Heatmap for Target & Aggregated Drivers

Figure 1 presents the correlation heatmap between the target variables and aggregated marketing drivers. `FIRST_PURCHASES` and `ALL_PURCHASES` show a very strong correlation (0.95), indicating that both targets capture closely related purchasing behaviour. In contrast, total paid spend shows only a weak direct correlation with purchases (≈ 0.02), suggesting that marketing spend alone does not linearly explain daily purchase variation. Engagement variables, such as clicks and impressions, exhibit stronger correlations (up to 0.90), suggesting potential multicollinearity among exposure-related features. This observation supports the use of regularised models and time-aware features during modelling.



Figure 2. Time-series Behaviour of Key Variables

Figure 2 illustrates the time-series behaviour of key variables. Daily purchases, paid spend, and non-paid clicks show substantial fluctuations and several extreme spikes, likely corresponding to major marketing campaigns or seasonal events. These patterns highlight the dynamic nature of marketing performance and suggest that lagged or temporal features may be necessary to capture delayed marketing effects.

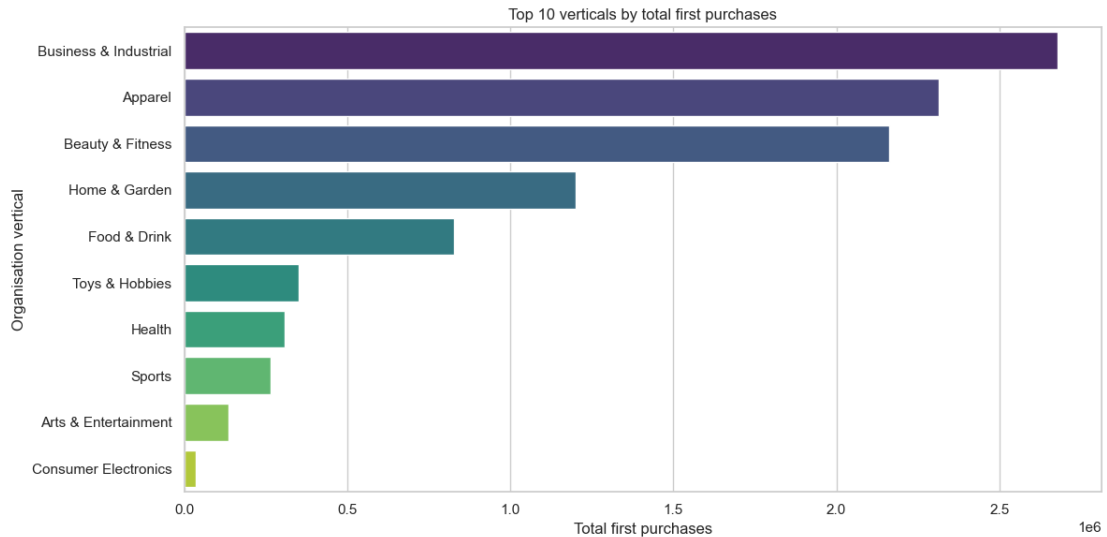


Figure 3. Top 10 Verticals by Total First Purchases

Figure 3 shows the distribution of total first purchases across organisational verticals. Business & Industrial, Apparel, and Beauty & Fitness dominate total purchases, while several other verticals contribute relatively smaller volumes. This uneven distribution indicates that purchasing behaviour differs across categories and suggests that vertical-level differences may influence overall marketing performance.

Agent tooling (Codex) was used to generate initial visualisations and summaries, while all plots and interpretations were manually verified to ensure correctness.

4. Data Preparation

Data preparation was conducted to ensure that the dataset was suitable for predictive modelling while maintaining reproducibility and preventing data leakage. The preprocessing workflow followed a structured pipeline that included data cleaning, feature construction, and a time-aware dataset split.

First, media variables with missing values were treated as periods with no advertising activity and were replaced with zeros. Aggregated marketing features were then created, including total paid spend, total paid clicks, and total paid impressions, which summarise overall marketing exposure across channels. These aggregated variables

help capture the broader relationship between marketing activity and purchasing outcomes.

Because the dataset is time-series based, a chronological split was used. (Table 1) The training set covers August 2021 to July 2023 (701 observations), the validation set covers July 2023 to December 2023 (150 observations), and the test set covers December 2023 to May 2024 (151 observations). This preserves temporal order and prevents information leakage.

	split	rows	date_min	date_max
0	train	701	2021-08-23	2023-07-24
1	validation	150	2023-07-25	2023-12-21
2	test	151	2023-12-22	2024-05-20

Table 1. Train / Validation / Test Sets

Basic data validation checks were also performed to confirm that no missing values remained after preprocessing and that key variables were within reasonable ranges. Agent tooling (Codex) was used to assist in generating parts of the preprocessing pipeline, including feature engineering suggestions. However, all proposed steps were manually reviewed and validated to ensure that they were appropriate for the dataset and consistent with the modelling objectives.

5. Model Development

The model development stage focuses on building and comparing several predictive models to estimate purchasing outcomes from historical marketing activity. Since the target variable represents purchase volume, the task is formulated as a regression problem.

A **seasonal naive model** was first established as a baseline. This simple benchmark assumes that future values follow historical seasonal patterns and provides a reference point for evaluating more sophisticated models.

Initially, Codex suggested tree-based machine learning models, including **Random**

Forest and **Histogram Gradient Boosting**. These models were implemented and evaluated using validation data; however, their predictive performance was relatively poor for this time-series marketing dataset. The results suggested that these models struggled to capture the underlying temporal dynamics and multicollinearity among marketing exposure variables.

Based on this observation, additional research was conducted to identify modelling approaches commonly used in Marketing Mix Modelling (MMM). Two relevant approaches were identified: **ElasticNet regression**, which combines L1 and L2 regularisation to address multicollinearity and feature selection, and **Google Meridian**, a Bayesian MMM framework. However, Meridian could not be used due to compatibility issues with Python 3.13. As an alternative, a **Bootstrapped Ridge regression** approach was implemented to improve model stability and robustness.

The final model comparison, therefore, included four models: a **seasonal naive baseline**, **Ridge regression**, **ElasticNet regression**, and **Bootstrapped Ridge regression**. Model performance was evaluated using validation data and standard regression metrics (MAE, RMSE, and MAPE). The regularised regression approaches demonstrated more stable and reliable predictive performance, supporting their suitability for modelling marketing response data.

6. Model Evaluation and Tuning

Model evaluation and tuning ensured that the selected models generated consistent, robust predictions. Due to the dataset's chronological structure, a time-aware data split was applied.

A chronological split partitioned the time-series dataset into training, validation, and test sets, preserving temporal order and preventing information leakage. The validation set enabled model comparison and hyperparameter tuning using out-of-sample performance.

Model performance was assessed with three regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error

(MAPE). These metrics gauge prediction accuracy, error size, and relative error. Four models were compared: a seasonal naive baseline, Ridge regression, ElasticNet regression, and Bootstrapped Ridge regression. The analysis aimed to identify which model predicted the target variable most accurately on the validation and test sets.

	model	is_ml_model	MAE	RMSE	MAPE	R2_log_scale
3	bootstrapped_ridge	True	2,051.7917	2,912.0379	0.1622	0.8059
1	ridge	True	2,067.5937	2,934.6020	0.1620	0.8033
0	seasonal_naive_lag_7	False	2,470.8109	3,435.3096	0.2241	0.6963
2	elastic_net	True	3,885.6324	4,436.0783	0.4149	0.3776

Table 2. Evaluation Result of 4 Models

The regularised regression models significantly outperformed the baseline. As shown in Table 2, the Bootstrapped Ridge model excelled, delivering the lowest MAE (2051.79), lowest RMSE (2912.04), and strong explanatory power ($R^2 = 0.8059$ on the log scale). Ridge regression results were similar, while ElasticNet performed worse, indicating that excessive regularisation reduced predictive performance.

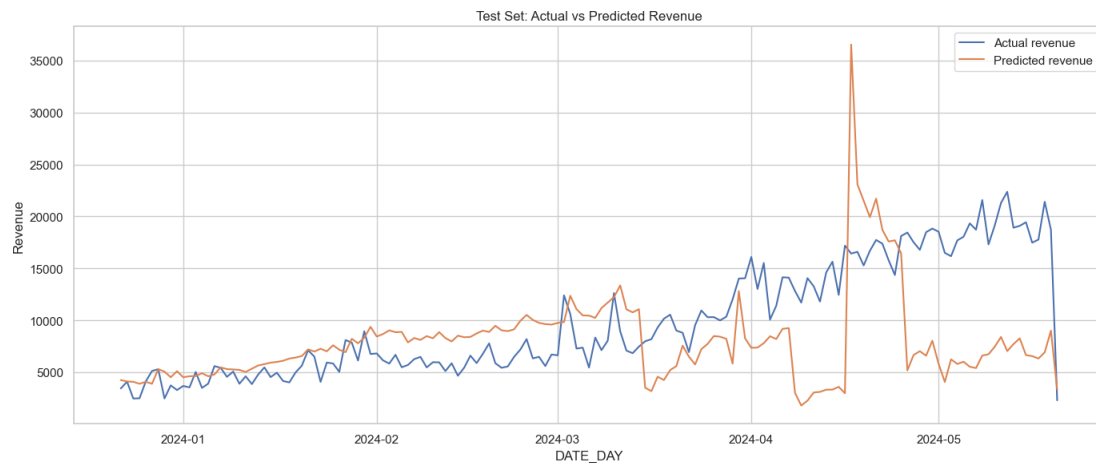


Figure 4. Actual vs Predicted Revenue

Figure 4 compares the actual revenue with predictions generated by the **Bootstrapped Ridge model** on the test dataset. The predicted series generally follows the overall trend of the observed revenue, indicating that the model captures the main dynamics of

purchasing behaviour. However, noticeable deviations occur during periods of sharp revenue spikes, where the model either overestimates or underestimates extreme values.

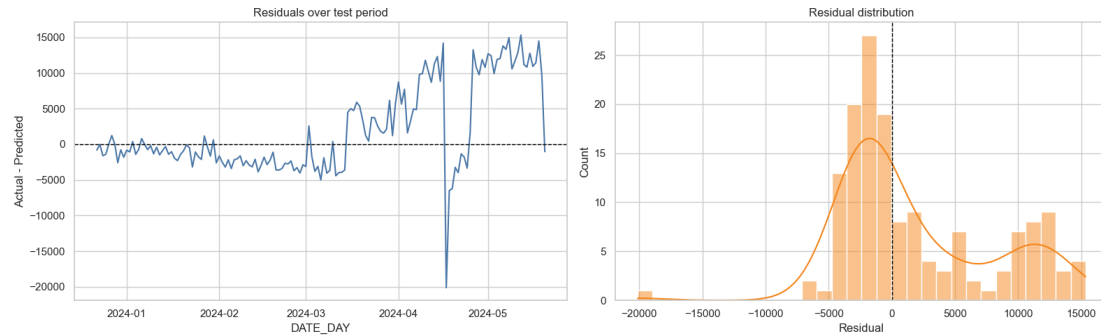


Figure 5. Residual Error Analysis

Residual error analysis shown in Figure 5 was conducted to further evaluate the **Bootstrapped Ridge model**. Most residuals fluctuate around zero, indicating that the model captures the general trend of purchasing behaviour. However, several large residual spikes occur during major campaign periods, suggesting that sudden marketing shocks remain difficult for the model to fully capture.

During development, an agent-suggested modelling approach introduced a potential issue. The agent (Codex) initially recommended tree-based models such as Random Forest and Histogram Gradient Boosting. After implementing and evaluating these models, experimental results showed that their predictive performance was significantly worse than that of the regression-based approaches. As a result, these models were rejected. The strategy shifted to regularised regression approaches common in marketing mix modelling, ensuring empirical, validated model selection.

7. Final Model and Solution

The **Bootstrapped Ridge regression model** was chosen for its top predictive performance and stable results. It was used to analyse the marginal effectiveness of

marketing channels and simulate budget allocations.

Item	Description
Model	Bootstrapped Ridge Regression
Task	Revenue prediction for marketing performance analysis
Target variable	Daily revenue / first purchases
Training data	Aug 2021 – Jul 2023
Validation data	Jul 2023 – Dec 2023
Test data	Dec 2023 – May 2024
Key inputs	Aggregated marketing exposure variables (spend, clicks, impressions)
Evaluation metrics	MAE, RMSE, MAPE, R ² (log scale)
Best performance	MAE $\approx$ 2051.8, RMSE $\approx$ 2912.0
Primary use	High-level marketing performance analysis and budget allocation insights
Not intended for	Precise short-term forecasting or capturing sudden campaign shocks

Table 3. Model Card of Bootstrapped Ridge Regression

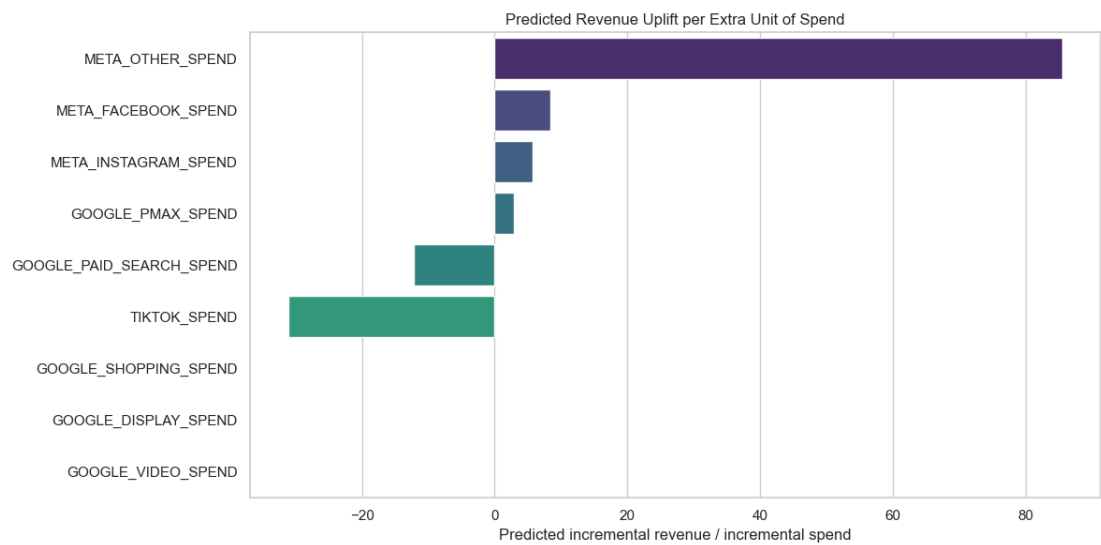


Figure 6. Predicted Revenue Uplift per Extra Unit of Spend

The uplift-per-extra-spend analysis (Figure 6) indicates that META_OTHER_SPEND has the highest predicted marginal return (≈ 85.47), substantially higher than other channels such as META_FACEBOOK_SPEND, META_INSTAGRAM_SPEND, and GOOGLE_PMAX_SPEND.

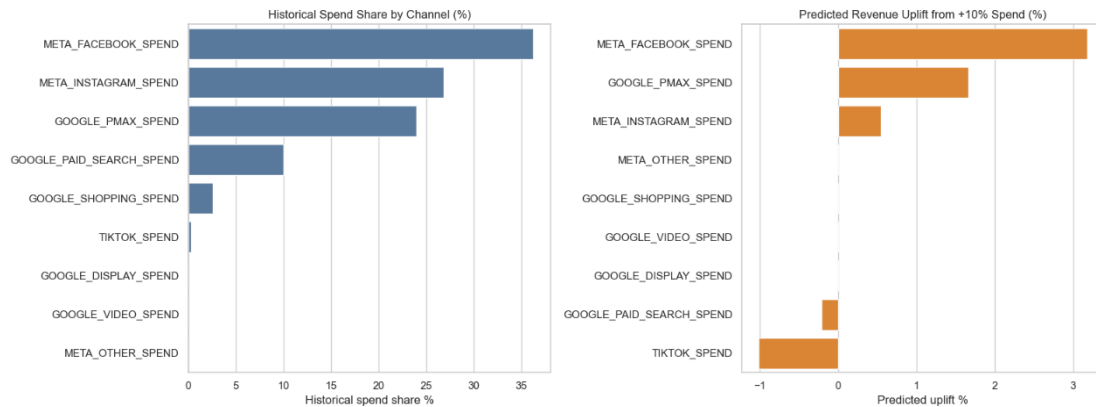


Figure 7. Historical Spend Share & Predicted Revenue Uplift from +10% Spend (%)

In the +10% spend simulation (Figure 7), this channel also produces a positive predicted revenue uplift, while GOOGLE_PAID_SEARCH_SPEND and TIKTOK_SPEND show negative responses, suggesting that additional spending in these channels may not increase revenue under the current model configuration. Given that META_OTHER_SPEND currently receives almost no historical budget, the model identifies it as potentially under-invested relative to its predicted response.

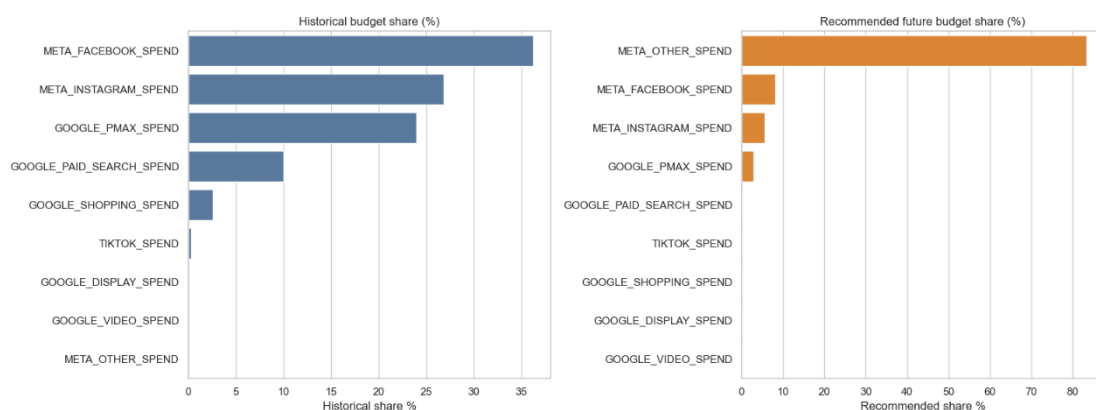


Figure 8. Historical Budget Share & Recommended Future Budget Share (%)

	channel	historical_spend_share_pct	uplift_per_extra_spend	predicted_revenue_uplift	positive_efficiency	recommended_share_pct	recommended_budget_amount
0	META_OTHER_SPEND	0.0065	85.4744	0.2173	85.4744	83.4501	83,450.1384
1	META_FACEBOOK_SPEND	36.2910	8.3475	105.8813	8.3475	8.1498	8,149.7963
2	META_INSTAGRAM_SPEND	26.8331	5.7028	18.3085	5.7028	5.5677	5,567.7058
3	GOOGLE_PMAX_SPEND	24.0107	2.9011	55.1670	2.9011	2.8324	2,832.3596
4	GOOGLE_PAID_SEARCH_SPEND	10.0117	-12.2432	-7.1133	0.0000	0.0000	0.0000
5	TIKTOK_SPEND	0.2634	-31.0220	-33.8760	0.0000	0.0000	0.0000
6	GOOGLE_SHOPPING_SPEND	2.5566	0.0000	0.0000	0.0000	0.0000	0.0000
7	GOOGLE_DISPLAY_SPEND	0.0180	0.0000	0.0000	0.0000	0.0000	0.0000
8	GOOGLE_VIDEO_SPEND	0.0090	0.0000	0.0000	0.0000	0.0000	0.0000

Table 4. Recommended Allocation of Additional Marketing Budget

For the final solution, based on the optimisation simulation (Figure 8 & Table 4), **the recommended allocation of additional marketing budget prioritises META_OTHER_SPEND (≈83%), followed by smaller allocations to META_FACEBOOK_SPEND (≈8%), META_INSTAGRAM_SPEND (≈6%), and GOOGLE_PMAX_SPEND (≈3%),** while other channels receive near-zero allocation due to weak or negative predicted marginal effects. These results suggest that reallocating incremental budget toward channels with stronger marginal efficiency could improve overall revenue performance.

8. Limitations and Future Work

Despite providing useful insights, several limitations should be acknowledged. The recommended budget allocation should be interpreted as a **model-based directional signal rather than a definitive optimisation result**, and it should be validated through controlled experiments or broader marketing mix modelling analysis before large-scale budget reallocation.

In addition, the dataset only includes observed marketing activity variables and does not capture external factors such as macroeconomic conditions, competitor behaviour, or offline campaigns. These missing influences may partly explain the large residual errors observed during periods of sudden revenue spikes. Some marketing channels also had high rates of missingness and required aggregation during preprocessing, potentially reducing channel-level interpretability.

Future work could incorporate additional contextual variables, model advertising carryover and saturation effects, and implement more advanced marketing mix modelling approaches to produce more robust budget allocation recommendations.

9. Appendix

a. Code Repository Link

<https://github.com/codeandworkhard/predictive-analytics-MMM>

b. Dataset Source

Anderson, Anthony (2024). Multi-Region Marketing Mix Modelling (MMM) Dataset for Several eCommerce Brands. figshare. Dataset.

<https://doi.org/10.6084/m9.figshare.25314841.v3>

c. Agent Usage Log

```
# Agent Usage Log

Use this file to document any AI or agent assistance referenced in the coursework brief.
Date ranges below are based on the chat timeline over the last 3 weeks.

| Date | Tool | Purpose | Prompt / Instruction Summary | What You Used in the Submission | What You Checked Manually |
| --- | --- | --- | --- | --- | --- |
| 2026-02-15 to 2026-02-16 | Codex | Project scaffolding | Asked to set up the assignment folder for an end-to-end ML project with "README" and "requirements.txt". | Repository structure, initial "README.md", and dependency list. | Confirmed files/folders exist and paths match local project. |
| 2026-02-16 to 2026-02-20 | Codex | Notebook build (Step 1) | Asked to create "01_data_cleaning_eda.ipynb" for data cleaning, processing, and EDA. | Data loading/cleaning pipeline, EDA sections, and cleaned dataset handoff for modelling. | Verified notebook runs and that cleaned outputs are generated. |
| 2026-02-20 | Codex | Environment/debug support | Reported "ModuleNotFoundError: No module named 'src'" in notebook imports. | Import-path fix for notebook execution context. | Re-ran import cell and checked notebook JSON validity. |
| 2026-02-20 to 2026-02-25 | Codex | Modelling notebook (Step 2) | Asked to implement revenue forecasting and channel recommendation workflow in notebook 02. | Time-aware split, feature engineering, model training, evaluation, and scenario analysis workflow. | Checked split logic, metric outputs, and plot generation. |
| 2026-02-25 to 2026-02-27 | Codex | Model configuration updates | Asked to replace "random_forest"/"hist_gradient_boosting" with ElasticNet MMM and Meridian, then replace Meridian with Bootstrapped Ridge due Python 3.13 compatibility. | Final model set includes ElasticNet and Bootstrapped Ridge MMM-style approach. | Confirmed code runs without Meridian dependency and model paths are correct. |
| 2026-02-27 to 2026-03-02 | Codex | Notebook QA and anonymity edits | Asked to add per-line comments, add interpretation blocks under figures/tables, and stop printing absolute local paths. | Expanded comments, added figure/table interpretation markdown, and anonymized path-print behavior. | Reviewed rendered outputs and checked no local absolute path is printed by notebook logic. |
| 2026-03-02 to 2026-03-05 | Codex | Workflow/evaluation enhancement | Asked to align with framing sequence and add model metrics plus residual/error-distribution diagnostics. | Added workflow sections, metric rationale ("RMSE" primary), and residual diagnostics in notebook 02. | Verified section order, chart outputs, and metric tables. |
| 2026-03-05 to 2026-03-07 | Codex | Budget allocation output | Asked to show recommended channel percentages for a specified future budget and include output/conclusion at notebook end. | Added "recommended_share_pct" and "recommended_budget_amount", allocation visuals/tables, and final channel split conclusion. | Executed notebook 02 and checked all channels display numeric percentages. |
| 2026-03-07 to 2026-03-08 | Codex | Iterative interpretation refinement | Repeatedly asked to replace generic interpretation text with chart-specific data analysis and fix markdown line-by-line formatting. | Updated multiple interpretation blocks in notebooks 01 and 02 with data-driven findings from each figure/table. | Confirmed markdown renders correctly (no literal '\n') and notebook JSON remains valid after edits. |
```

Table 5. Agent Usage Log Generated by Codex

(1) Project setup and workflow planning

The agent was used to suggest an initial repository structure and notebook workflow for the machine learning project. The generated structure included separate notebooks for data preparation and modelling. The structure was reviewed manually and adapted to match the coursework requirements.

(2) Model selection discussion

The agent suggested tree-based models such as Random Forest and Histogram

Gradient Boosting for revenue prediction. After testing these models on the validation set, their performance was worse than Ridge-based models. Therefore, the suggestion was rejected and Ridge-based approaches were prioritised.

(3) Model evaluation guidance

The agent recommended performing residual diagnostics to better understand prediction errors. Based on this suggestion, residual plots and residual distribution charts were added to the modelling notebook to analyse error patterns across the test period.

(4) Budget allocation interpretation

The agent helped interpret the uplift-per-extra-spend metric as a measure of marginal channel efficiency. This interpretation was used to produce marginal efficiency charts and simulate potential marketing budget allocation scenarios.

(5) Notebook explanation refinement

The agent suggested adding interpretation blocks under figures and tables to clarify modelling results. These explanations were manually refined to include chart-specific insights rather than generic descriptions.

d. Decision Register

```
## Verification decisions (accepted / modified / rejected)
| Date range | Agent contribution | Your verification decision | How it changed after your review |
| --- | --- | --- | --- |
| 2026-02-25 to 2026-02-27 | Proposed model set update including ElasticNet + Meridian replacement of tree models. | Modified | You changed the request because Meridian was not feasible in your Python 3.13 environment; final notebook uses ElasticNet + Bootstrapped Ridge instead. |
| 2026-02-27 to 2026-03-02 | Added long markdown guidance block for coursework write-up. | Rejected | You asked to remove that section; it was deleted from notebook 02. |
| 2026-03-02 to 2026-03-05 | Added workflow wording in notebook 01 that included problem framing. | Modified | You clarified notebook 01 should focus only on cleaning/processing/EDA; workflow text was revised and problem-framing wording was removed. |
| 2026-03-05 to 2026-03-07 | Added budget allocation section with channel percentages and recommendation output in notebook 02. | Accepted | You kept the section and requested further refinement (full channel percentages + conclusion details), which was implemented. |
| 2026-03-07 to 2026-03-10 | Added figure/table interpretation blocks. | Modified | You repeatedly asked to replace generic comments with chart-specific data analysis and line-by-line markdown formatting; interpretations were rewritten accordingly. |
```

Table 6. Verification Decisions Log (Accepted / Modified / Rejected) Generated by Codex

Step	Agent Suggestion	Decision	Verification
Project setup	Suggested repository structure and notebook workflow for the ML project	Accepted	Confirmed folder structure, README, and dependencies match the project setup

Data preparation	Suggested building a data cleaning and EDA pipeline in notebook 01	Accepted	Verified cleaned dataset outputs and checked preprocessing logic
Model selection	Suggested Random Forest and Histogram Gradient Boosting models	Rejected	Validation performance was worse than Ridge-based models
Final model	Suggested ElasticNet MMM approach	Modified	Switched to Bootstrapped Ridge regression for better stability and compatibility
Evaluation	Suggested residual diagnostics and error distribution analysis	Accepted	Added residual plots and residual distribution charts to evaluate model errors
Budget allocation	Suggested interpreting uplift-per-extra-spend as marginal efficiency	Accepted	Used to generate channel marginal efficiency analysis and recommendation charts
Output interpretation	Suggested adding explanation blocks under figures and tables	Modified	Replaced generic explanations with chart-specific interpretation and analysis

Table 7. Agent Suggestions & Verification Decisions (Accepted / Modified / Rejected) Generated by Codex